2024-2025

S4 MATH CP

PAPER I

CUHKFAA CHAN CHUN HA SECONDARY SCHOOL

2024-2025 Secondary Four First Term Examination

MATHEMATICS Compulsory Part

PAPER I

Question-Answer Book

03/01/2025

8:15 am – 10:30 am (2¼ hours)

This paper must be answered in English

INSTRUCTIONS

1. After the announcement of the start of the examination, you should first write your Name, Class and Class number in the spaces provided on Page 1.

2. This paper consists of THREE sections, A(1), A(2) and B.

3. Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

4. Graph paper and supplementary answer sheets will be supplied on request. Write your Name, Class and Class number and mark the question number box on each sheet, and fasten them with string INSIDE this book.

5. Unless otherwise specified, all working must be clearly shown.

6. Unless otherwise specified, numerical answers should be either exact or correct to 3 significant figures.

7. The diagrams in this paper are not necessarily drawn to scale.

Name: _____

Class: _____( )

	Marker Use Only
Question	Mark
1	
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Total			

SECTION A(1) (35 marks)

1. Simplify $\dfrac{ab^{-5}}{(a^2b^{-4})^{-4}}$ and express your answer with positive indices. (3 marks)

2. Factorize

   (a) $9m^2 - 10m + 1$,

   (b) $27mn - 3n - 9m^2 + 10m - 1$. (4 marks)

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3. Make y the subject of the formula $k - y = \frac{5x + ky}{2}$. (3 marks)

4. **(a)** Round off 0.4991 to the nearest integer.
   (b) Round down 0.4991 to 2 significant figures.
   (c) Round up 0.4991 to 3 decimal places. (3 marks)

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5. The coordinates of the points P and Q are (–5, 7) and (4, –8) respectively. P is rotated clockwise about O through 90° to P', where O is the origin. Q is translated vertically upwards by 18 units to Q'.

(a) Write down the coordinates of P′ and Q′.

(b) Is PQ parallel to P′Q′ ? Explain your answer. (4 marks)

6. Three years ago, the ages of Alice and John were in the ratio 2 : 3. Now, the ratio of the ages of Alice and John will become 3 : 4. Find the present ages of Alice and John. (4 marks)

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7. The length of a piece of thin metal wire is measured as 6.1m correct to the nearest 0.1m. Let x m be the actual length of the wire.

   (a) Find the range of the values of x.

   (b) Is it possible to cut this metal wire into 32 pieces of shorted metal wires, with each length measured as 20 cm correct to the nearest cm? Explain your answer. (4 marks)

8. The cost price of a pencil case is $46. The pencil case is now sold and the profit per cent is 50%.

   (a) Find the selling price of the pencil case.

   (b) If the pencil case is sold at a discount of 40% on its marked price, find the marked price of the pencil case. (4 marks)

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9. If the y-intercept of the straight line L: $x - Ay + 3 = 0$ is 9, find

(a) the value of A,

(b) the x-intercept of L. (3 marks)

10. Consider the graph of the quadratic function $y = 4x^2 - 24x + 11$.

(a) Find the x-intercepts of the graph. (1 mark)

(b) Find the equation of the axis of symmetry of the graph. (1 mark)

(c) Find the coordinates of the vertex of the graph. (1 mark)

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SECTION A(2) (35 marks)

11. It is given that $f(x) = x^2 + ax$, where $a \neq 0$. If $f(a) = 2a$, find

(a) the value of a, (2 marks)

(b) $f(-1)$. (2 marks)

12. It is given that the three straight lines L_1: $y = 2x - 5$, L_2: $3x - 2y - 6 = 0$ and L_3: $x - ky = 0$ intersect at a point A.

(a) Find the coordinates of the intersection of L_1 and L_2. (2 marks)

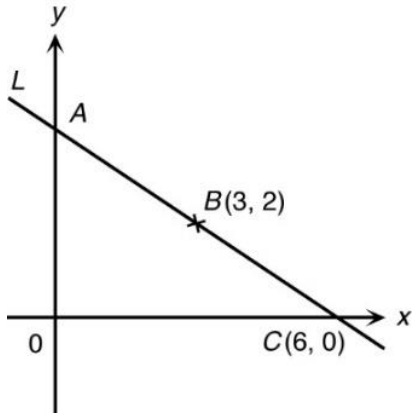
(b) Find the values of k. (2 marks)

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13. In the figure, the straight line L cuts the y-axis at A. $B(3, 2)$ and $C(6, 0)$ are two points on L.



(a) Find the equation of L. (3 marks)

(b) **(i)** Find the coordinates of A.

(ii) Hence, show that B is the mid-point of AC. (3 marks)

(c) Find the perpendicular bisector of AC. (2 marks)

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14. Given that $\sin\theta = \frac{2}{5}$ and $\cos\theta < 0$, find the value of $\cos\theta - \tan\theta$. Give your answer in the simplest surd form. (5 marks)

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15. Let $f(x) = 6x^3 - 17x^2 - 5x + 6$.

(a) Show that $x - 3$ is a factor of $f(x)$. (2 marks)

(b) Hence, by using long division, factorize the polynomial $f(x)$. (3 marks)

(c) Someone claims that all the roots of the equation $f(x) = 0$ are rational numbers. Do you agree? Explain your answer. (2 marks)

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16. The graph of $y = 2(x - h)^2 - 8$ passes through (1, 10), where h is a constant.

(a) Find the minimum value of y. (1 mark)

(b) Find all possible values of h. (2 marks)

(c) For each value of h, find the x-intercept(s) of the graph. (4 marks)

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SECTION B (35 marks)

17. **(a)** Prove that $\sin x - 3\cos x + \tan x - 3 = (\cos x + 1)(\tan x - 3)$. (2 marks)

(b) Hence, or otherwise, solve the equation $\sin x - 3\cos x + \tan x - 3 = 0$ for $180° \le x \le 360°$.

(4 marks)

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18. The total cost \$$C$ of making n balls can be expressed as $C = k + 35n$, where k is a constant. Suppose that the total cost of making 60 balls is \$2700.

(a) Find the value of k. (2 marks)

(b) If 80 balls are made, find the average cost of making a ball. (2 marks)

(c) The marked price of a ball is \$90. 80 balls are now made, 85% of them are sold at 15% discount and the rest are sold at half price. Find the percentage profit of making and selling these 80 balls. (4 marks)

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19. It is given that $f(x) = 10x^3 + ax^2 + bx - 24$. When $f(x)$ is divided by $x - 1$ and $x - 2$, the remainders are -26 and 54 respectively.

(a) Find the values of a and b. (5 marks)

(b) Emily claims that $f(x)$ is divisible by $2x - 3$. Is she correct? Explain your answer. (2 marks)

(c) Simplify $\frac{5x^2 + 13x + 8}{f(x)} \times \frac{(2x-3)^2}{5x+8}$. (3 marks)

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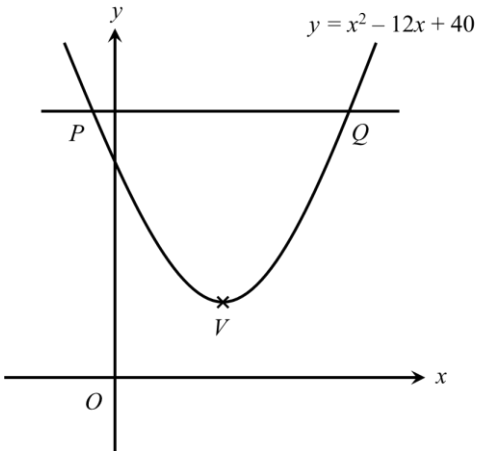
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20. The figure shows the graph of $y = x^2 - 12x + 40$. V is the vertex of the graph. It is given that the y-coordinate of V is 4. A horizontal line cuts the graph at P and Q. Let m be the x-coordinate of Q.



(a) Show that $x - 15$ is a factor of $x^3 - 20x^2 + 94x - 285$. (2 marks)

(b) Find the vertex of the graph. (2 marks)

(c) Hence, **(i)** express the length of PQ in terms of m.

(ii) find the area of $\triangle PVQ$ in terms of m.

(iii) W is a point on the rectangular coordinate plane such that $PVQW$ is a rhombus. If the area of $PVQW$ is $(4m^2 + 28m + 138)$ sq. units, find the value of m.

(7 marks)

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